Lab 3

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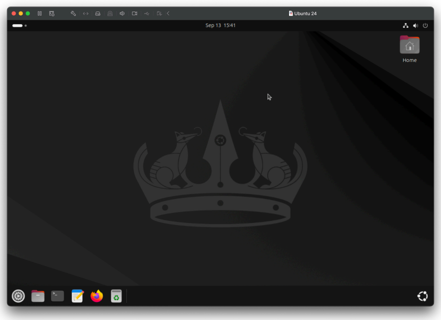
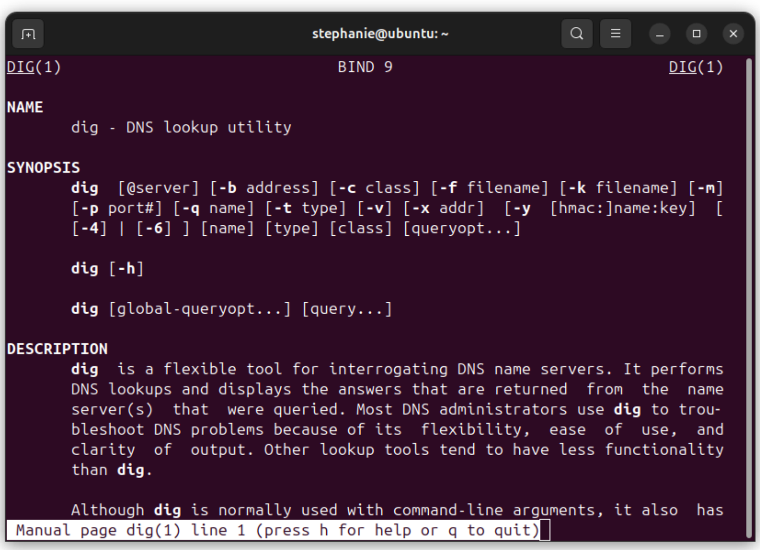
Using a Linux VM



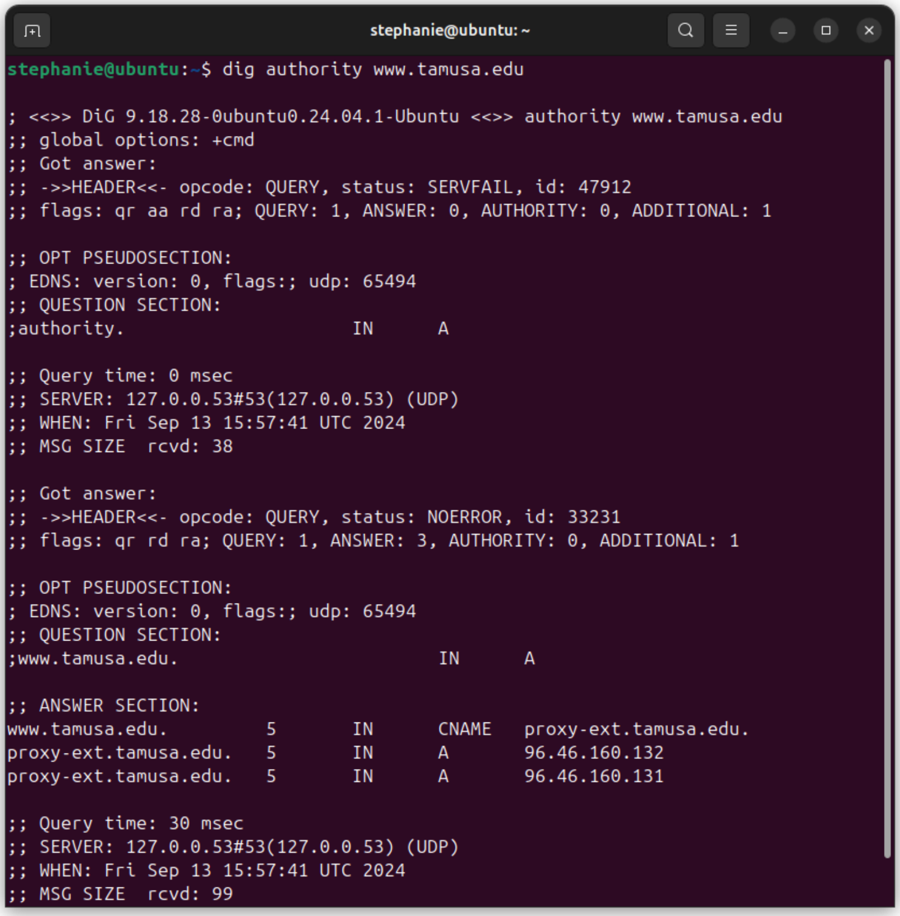
Since we needed Linux for this lab, I used VMware Fusion on my Mac to add and configure an Ubuntu 24 VM.

The "dig" command



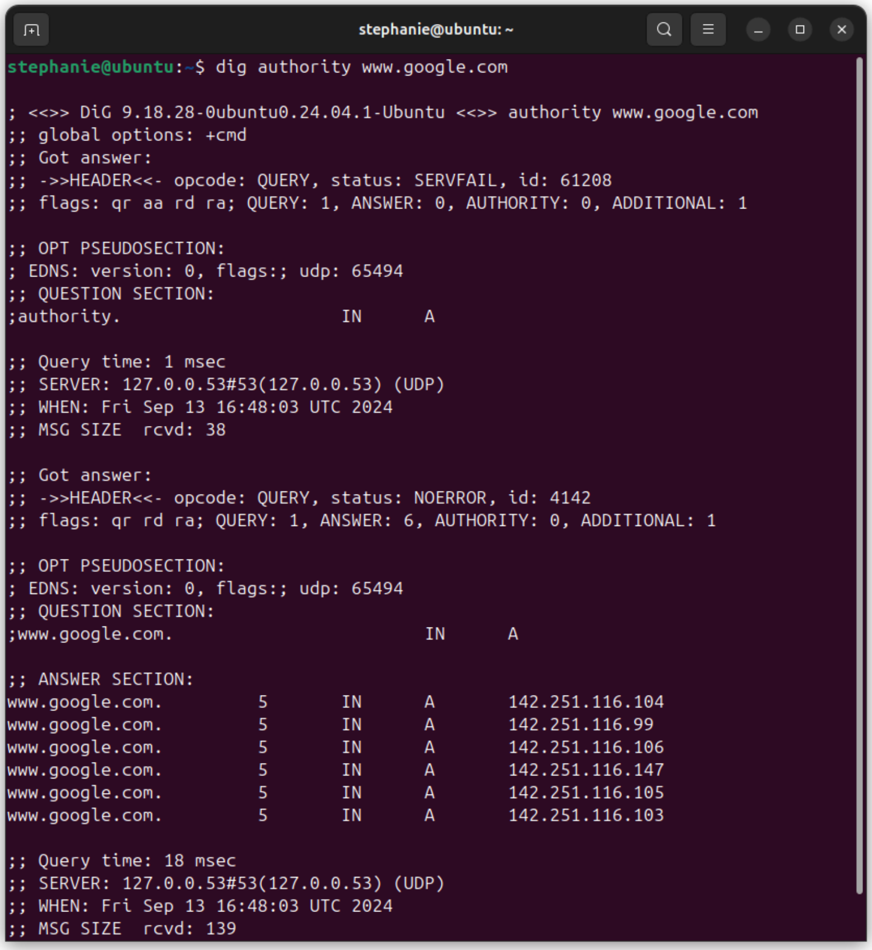
The **d**omain **i**nformation **g**roper command is used to gather DNS information.

Using “dig authority” on tamusa



The answer section is what provides the information we queried. In this case, the first query had a status of “SERVFAIL” indicating there was a failure resolving the DNS query. The query was made to “127.0.0.53”, the DNS resolver. The second query had a status of “NOERROR”, meaning the DNS query was successful. The answer section reflects this. The first column in this section corresponds to the domain names, notice the 4th column, it says “CNAME” for “www.tamusa.edu”, this shows it’s an alias pointing to “proxy-ext.tamusa.edu”. The “A” for both is the “Address” record, which resolves to 2 IPs. The “5” signifies the TTL (in seconds), which defines how long the DNS record is cached. The “IN” simply indicates the DNS record class is Internet.

Using “dig authority” on google



```
stephanie@ubuntu:~$ dig authority www.google.com

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> authority www.google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 61208
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;authority.                     IN      A

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 16:48:03 UTC 2024
;; MSG SIZE  rcvd: 38

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 4142
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.google.com.                IN      A

;; ANSWER SECTION:
www.google.com.         5       IN      A       142.251.116.104
www.google.com.         5       IN      A       142.251.116.99
www.google.com.         5       IN      A       142.251.116.106
www.google.com.         5       IN      A       142.251.116.147
www.google.com.         5       IN      A       142.251.116.105
www.google.com.         5       IN      A       142.251.116.103

;; Query time: 18 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 16:48:03 UTC 2024
;; MSG SIZE  rcvd: 139
```

The first query had the same status as when I used it on “tamusa”. The main difference here is that “www.google.com” has multiple “A” records (IPv4 addresses), which is normal with high traffic websites for load balancing. Meaning anytime you query the site, the DNS can return any of the IPs. Additionally, the query time for google was quicker, at 18ms, when compared to 30ms for “tamusa”.

I noticed that if the first query was successful for both “tamusa” and “google”, there would’ve been an “SOA”, “Start of Authority”, record which would display “a.root-servers.net. nstld.verisign-grs.com.” under the authority section.

Using “dig nssearch” on tamusa, google and facebook

```
stephanie@ubuntu:~$ dig nssearch www.tamusa.edu

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> nssearch www.tamusa.edu
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 44054
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;nssearch.                      IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 17:17:49 UTC 2024
;; MSG SIZE  rcvd: 37

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 26950
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.tamusa.edu.                        IN      A

;; ANSWER SECTION:
www.tamusa.edu.         5       IN      CNAME   proxy-ext.tamusa.edu.
proxy-ext.tamusa.edu.   5       IN      A       96.46.160.131
proxy-ext.tamusa.edu.   5       IN      A       96.46.160.132

;; Query time: 120 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 17:17:49 UTC 2024
;; MSG SIZE  rcvd: 99
```

```
stephanie@ubuntu:~$ dig nssearch www.google.com

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> nssearch www.google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 53181
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;nssearch.                      IN      A

;; Query time: 2 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 17:20:55 UTC 2024
;; MSG SIZE  rcvd: 37

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8119
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.google.com.                        IN      A

;; ANSWER SECTION:
www.google.com.         5       IN      A       142.251.116.104
www.google.com.         5       IN      A       142.251.116.99
www.google.com.         5       IN      A       142.251.116.106
www.google.com.         5       IN      A       142.251.116.103
www.google.com.         5       IN      A       142.251.116.147
www.google.com.         5       IN      A       142.251.116.105

;; Query time: 17 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri Sep 13 17:20:55 UTC 2024
;; MSG SIZE  rcvd: 139
```

I got the “SERVFAIL” status when trying “dig nssearch”. From the sample, I know the command would have given similar SOA information for each.

Trying on Mac

For “www.tamusa.edu”:

```
;; QUESTION SECTION:
;nssearch.                      IN      A

;; AUTHORITY SECTION:
.                       86399   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091300 1800 900 604800 86400
```

For “www.google.com”:

```
;; QUESTION SECTION:
;nssearch.                      IN      A

;; AUTHORITY SECTION:
.                       86390   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091300 1800 900 604800 86400
```

For “www.facebook.com”:

```
;; QUESTION SECTION:
;nssearch.                      IN      A

;; AUTHORITY SECTION:
.                       86395   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400
```

Using “dig additional” on facebook and tamusa

```
stephanie@ubuntu:~$ dig additional www.tamusa.edu

; <<>> DiG 9.18.28-0ubuntu0.24.04.1-Ubuntu <<>> additional www.tamusa.edu
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 2531
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
```

Notice I got the “SERVFAIL” status again. I decided to continue on Mac.

```
> dig additional www.tamusa.edu

; <<>> DiG 9.10.6 <<>> additional www.tamusa.edu
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 7101
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;additional.                    IN      A

;; AUTHORITY SECTION:
.                       86391   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400

;; Query time: 37 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; WHEN: Fri Sep 13 12:59:10 CDT 2024
;; MSG SIZE  rcvd: 114

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 54233
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.tamusa.edu.                        IN      A

;; ANSWER SECTION:
www.tamusa.edu.         3167    IN      CNAME   proxy-ext.tamusa.edu.
proxy-ext.tamusa.edu.   125     IN      A       96.46.160.132
proxy-ext.tamusa.edu.   125     IN      A       96.46.160.131

;; Query time: 14 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; WHEN: Fri Sep 13 12:59:10 CDT 2024
;; MSG SIZE  rcvd: 99
```

```
> dig additional www.facebook.com

; <<>> DiG 9.10.6 <<>> additional www.facebook.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 38698
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;additional.                    IN      A

;; AUTHORITY SECTION:
.                       86399   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400

;; Query time: 22 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; WHEN: Fri Sep 13 13:07:54 CDT 2024
;; MSG SIZE  rcvd: 114

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 53039
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.facebook.com.              IN      A

;; ANSWER SECTION:
www.facebook.com.       2777    IN      CNAME   star-mini.c10r.facebook.com.
star-mini.c10r.facebook.com. 33 IN      A       157.240.25.35

;; Query time: 28 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; WHEN: Fri Sep 13 13:07:54 CDT 2024
;; MSG SIZE  rcvd: 90
```

The first queries returned the same SOA records. The only major difference I can spot is in the TTLs. For some reason, they are larger with this kind of query.

Using “dig nsid” on tamusa and facebook

For “www.tamusa.edu”:

```
;; QUESTION SECTION:
;nsid.                          IN      A

;; AUTHORITY SECTION:
.                       86260   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400
```

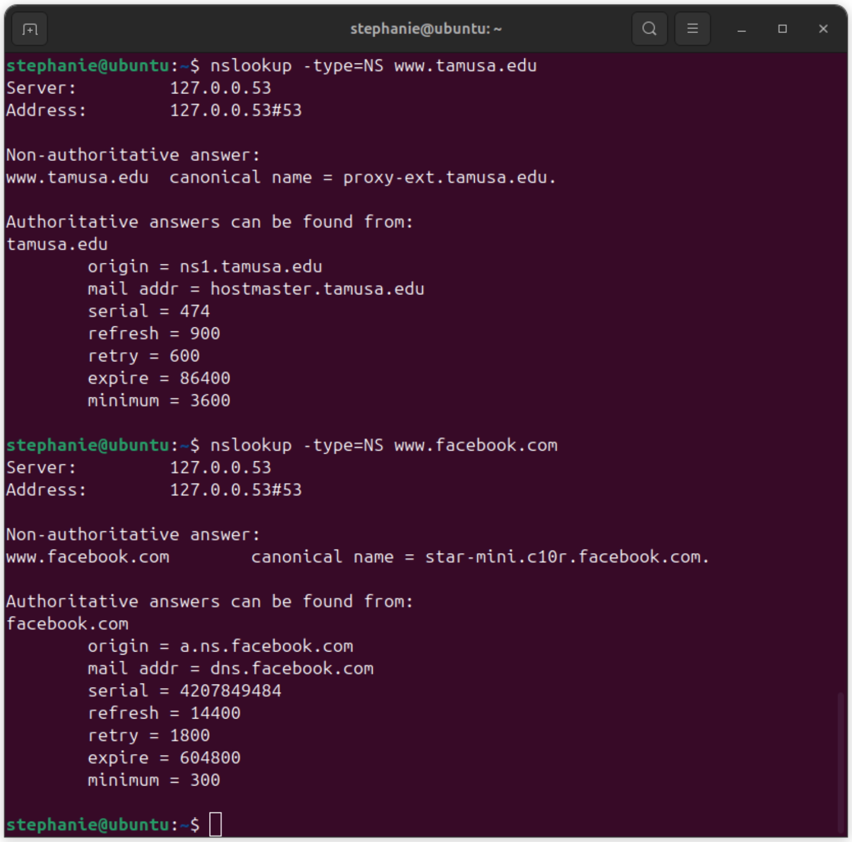
For “www.facebook.com”:

```
;; QUESTION SECTION:
;nsid.                          IN      A

;; AUTHORITY SECTION:
.                       86396   IN      SOA     a.root-servers.net. nstld.verisign-grs.com. 2024091301 1800 900 604800 86400
```

Again, the queries return the same SAO records and show no further changes.

Using “nslookup -type=NS” on tamusa and facebook

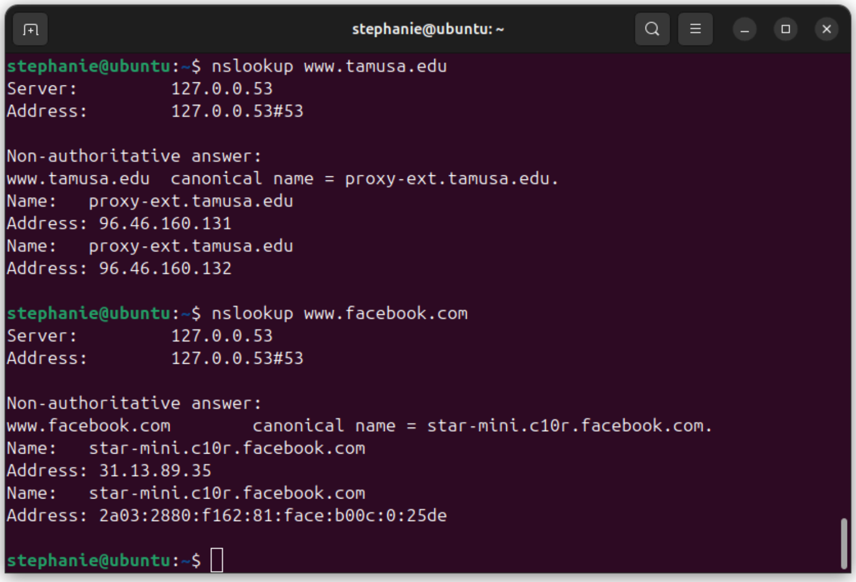


The “nslookup -type=NS” command outputs look simpler and easier to read. It returns the appropriate “127.0.0.53” DNS resolver server/address. This is because the command uses the “-type” flag with the value of “NS”, which queries for NS (Name Server) records specifically.

For “tamusa”, it displays the same “CNAME” information and shows where it points. It provides further information under the “Authoritative answers can be found from:” section.

For “facebook”, it displays the “CNAME” information which is pointing to “star-mini.c10r.facebook.com”. It also provides all the “Authoritative answers can be found from:” section detailing information like origin, mail addr, etc,..

Using “nslookup” on tamusa and facebook



Finally, the “nslookup” commands provide a similar view. Without the “-type” tag, it returns the default “Address” records. It can be noted that this time, the command provides the “Non-authoritative answer:”. It still displays the canonical names and where they point for both. However, this time, the addresses are also shown, one of them for facebook is in IPv6.

Summary

In this lab, I got to use the “dig” and “nslookup” commands to learn more about DNS. I encountered a problem with the “dig authority” and “dig nssearch” commands having a “SERVFAIL” status each time I ran them. I found success when running them on Mac. I believe it was likely due to the systems having different versions of “dig”. I had no problems when running the “nslookup” commands in the VM. I liked “nslookup” because it was simple. From this lab, I was able to learn about “canonical names”, and different types of DNS records like “Start of Authority”, “Address” and “Name Server”.